

KOYUK, AK, USA

WMO: 702223

Lat: 64.935N

Lon: 161.155W

Elev: 110

StdP: 14.64

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 26602

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.1	-19.9	-29.1	1.1	-15.2	-27.8	1.2	-15.4	25.0	1.9	23.1	1.0	2.7	50	0.774

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	73.3	57.5	70.1	57.1	65.7	55.4	60.6	69.6	58.5	66.7	56.5	64.0	7.1	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.5	68.2	63.6	54.4	63.3	61.7	52.4	58.8	59.7	26.8	69.9	25.3	66.6	24.0	63.8	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	18.7	17.0	-27.8	80.6	8.5	4.2	-34.0	83.6	-39.0	86.1	-43.8	88.4	-50.0	91.5		
(5)			-24.4	62.5	5.1	2.2	-28.1	64.1	-31.1	65.4	-34.0	66.6	-37.7	68.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.5	7.3	9.3	10.5	24.5	40.0	52.9	56.1	53.0	44.4	30.3	15.4	8.5
	DBStd	21.47	16.19	15.25	12.69	11.71	9.85	7.02	6.05	5.85	6.70	8.40	11.45	14.44
	HDD50	8002	1324	1141	1223	765	334	47	13	33	187	611	1038	1286
	HDD65	12986	1789	1561	1688	1215	776	368	284	373	617	1076	1488	1751
	CDD50	506	0	0	0	0	23	132	202	127	21	1	0	0
	CDD65	15	0	0	0	0	1	4	7	3	0	0	0	0
	CDH74	118	0	0	0	0	11	35	60	12	0	0	0	0
	CDH80	20	0	0	0	0	0	6	13	1	0	0	0	0
Wind	WSAvg	8.0	9.6	9.6	9.1	8.6	7.2	6.8	7.0	7.0	7.2	7.0	7.7	8.8
Precipitation	PrecAvg	20.6	1.3	1.3	1.1	0.9	1.0	1.3	2.5	3.3	2.6	2.1	1.7	1.5
	PrecMax	27.0	3.0	4.4	3.1	2.0	2.0	2.2	4.8	6.6	4.9	5.4	3.5	3.1
	PrecMin	13.1	0.3	0.3	0.2	0.3	0.3	0.6	0.4	0.9	1.1	0.6	0.4	0.6
	PrecStd	3.5	0.7	0.8	0.7	0.5	0.4	0.5	1.1	1.1	1.0	1.1	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	37.4	37.2	50.4	74.8	80.7	80.8	75.0	64.3	52.4	42.6	40.6
		MCWB	35.6	32.3	31.7	40.0	55.1	59.1	61.5	61.5	51.5	45.3	38.1	35.2
	2%	DB	36.3	35.8	34.4	46.0	64.5	72.8	73.4	70.2	60.6	46.0	37.0	36.3
		MCWB	32.5	31.5	30.0	37.0	50.6	55.9	58.5	57.7	50.0	41.5	33.4	32.1
	5%	DB	33.7	32.4	30.5	43.2	58.0	68.3	70.1	65.6	55.4	43.3	34.4	31.9
		MCWB	30.3	29.5	27.7	35.0	46.4	55.4	58.6	55.8	48.9	39.4	31.4	29.4
	10%	DB	28.4	28.5	27.9	39.0	54.1	63.9	65.4	62.8	53.9	41.0	32.1	27.9
		MCWB	26.1	25.7	24.8	32.5	44.4	52.7	56.8	54.5	48.3	37.9	30.1	25.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.4	33.4	32.8	40.4	55.4	61.2	64.5	63.0	53.4	45.8	38.9	36.2
		MCDB	41.8	37.1	36.4	48.6	73.1	70.4	75.5	73.9	59.2	50.8	41.8	38.3
	2%	WB	32.9	31.5	30.6	37.7	51.4	58.5	61.5	59.7	51.5	42.4	34.6	32.3
		MCDB	35.8	34.6	33.6	45.0	63.4	69.7	70.8	66.8	56.5	45.4	36.6	35.7
	5%	WB	30.0	29.5	28.1	35.5	48.1	56.0	59.2	57.4	50.3	40.4	31.9	29.3
		MCDB	32.4	32.2	30.7	42.1	56.6	65.7	67.5	63.3	55.2	43.4	33.9	31.5
	10%	WB	26.5	26.3	25.1	33.5	45.2	54.1	57.2	55.5	48.5	38.1	29.6	25.6
		MCDB	28.9	28.7	27.9	38.6	52.8	62.7	64.6	61.0	53.3	40.9	31.9	28.1
Mean Daily Temperature Range	5% DB	MDBR	9.3	10.7	11.9	13.1	14.3	14.0	11.4	11.5	11.1	8.7	8.4	9.0
		MCDBR	12.1	10.0	12.1	14.0	22.0	20.9	19.4	17.3	14.3	9.4	7.7	10.1
		MCWBR	10.3	8.2	9.8	9.5	13.1	11.3	10.6	9.7	9.1	7.5	7.2	8.6
	5% WB	MCDBR	11.8	10.3	11.8	12.7	20.3	17.5	16.7	13.8	12.5	9.1	7.9	9.6
		MCWBR	10.3	9.3	9.9	8.8	12.4	10.2	9.8	8.6	8.2	7.6	7.7	8.6
Clear-Sky Solar Irradiance	taub	0.178	0.255	0.269	0.342	0.359	0.356	0.363	0.384	0.313	0.273	N/A	0.181	
	taud	1.909	2.146	2.239	2.086	2.274	2.400	2.410	2.349	2.530	2.380	N/A	1.911	
	Ebn at Noon	111	204	260	258	267	272	266	246	247	201	N/A	31	
	Edh at Noon	7	18	26	39	36	32	31	30	20	14	N/A	3	
All-Sky Solar Radiation	RadAvg	46	243	802	1425	1661	1778	1437	1094	688	285	74	13	
	RadStd	6	29	67	87	158	162	103	133	71	36	10	1	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page